家禽腫瘤性疾病(續)

Neoplastic Diseases of Poultry

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Virus-induced transmissible neoplastic diseases

- Marek's disease 馬立克病
  - Marek's disease virus (MDV), herpesvirus
- Avian leukoses/sarcoma 家禽白血病
  - avian leukosis virus (ALV), retrovirus
- Reticuloendotheliosis 網狀內皮細胞增殖病
  - reticuloendotheliosis virus (REV), retrovirus

Avian Leukosis/Sarcoma

History

- The earliest reports: 1868, Roloff, “lymphosarcomatosis”
- On October 1, 1909, **Dr. Peyton Rous** removed a tumor from an English hen and injected a cell-free filtrate from the tumor into another healthy chicken, which later developed the same type of tumor

Cancer could be caused by a viral infection!

the Nobel Prize in 1966

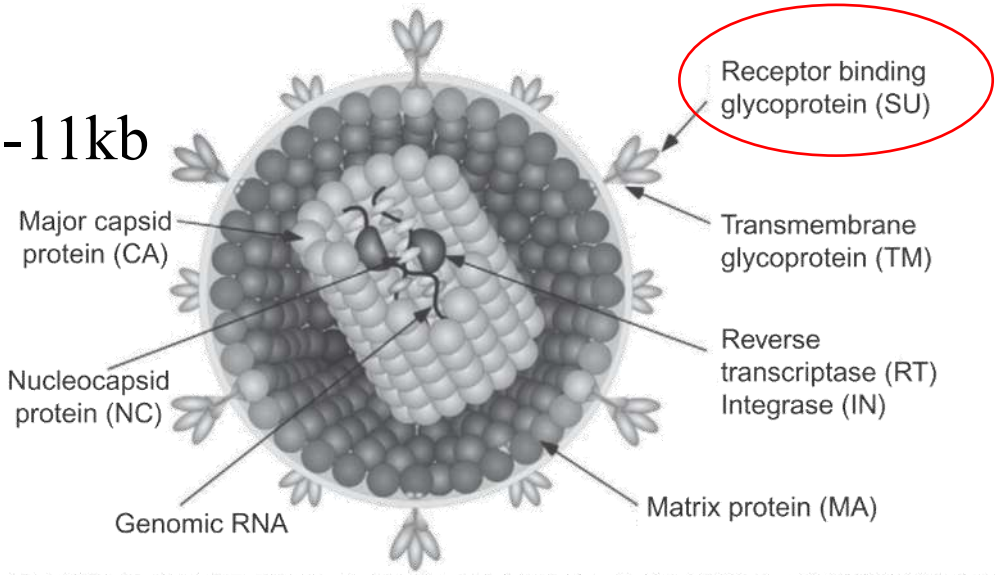


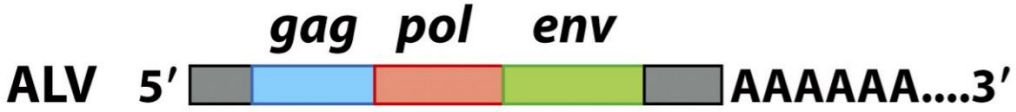


Virus type	Nucleic acid type	Virus classification of etiological agent	Neoplastic diseases
Retrovirus	RNA	Leukosis/sarcoma group	Leukoses Lymphoid leukosis Erythroblastosis Myeloloblastosis Sarcomas and other connective tissue tumors Fibrosarcoma, fibroma Myxosarcoma, myxoma Osteogenic sarcoma, osteoma Histiocytic sarcoma Related neoplasms Hemangioma Nephroblastoma Hepatocarcinoma Osteopetrosis
		Reticuloendotheliosis group	Reticuloendotheliosis Lymphoid leukosis
Herpesvirus	DNA	Marek's disease virus	Marek's disease

Avian leukosis virus (ALV)

- *Alpharetrovirus*
- (+) ssRNA virus, 7-11kb
- 80-100 nm
- enveloped



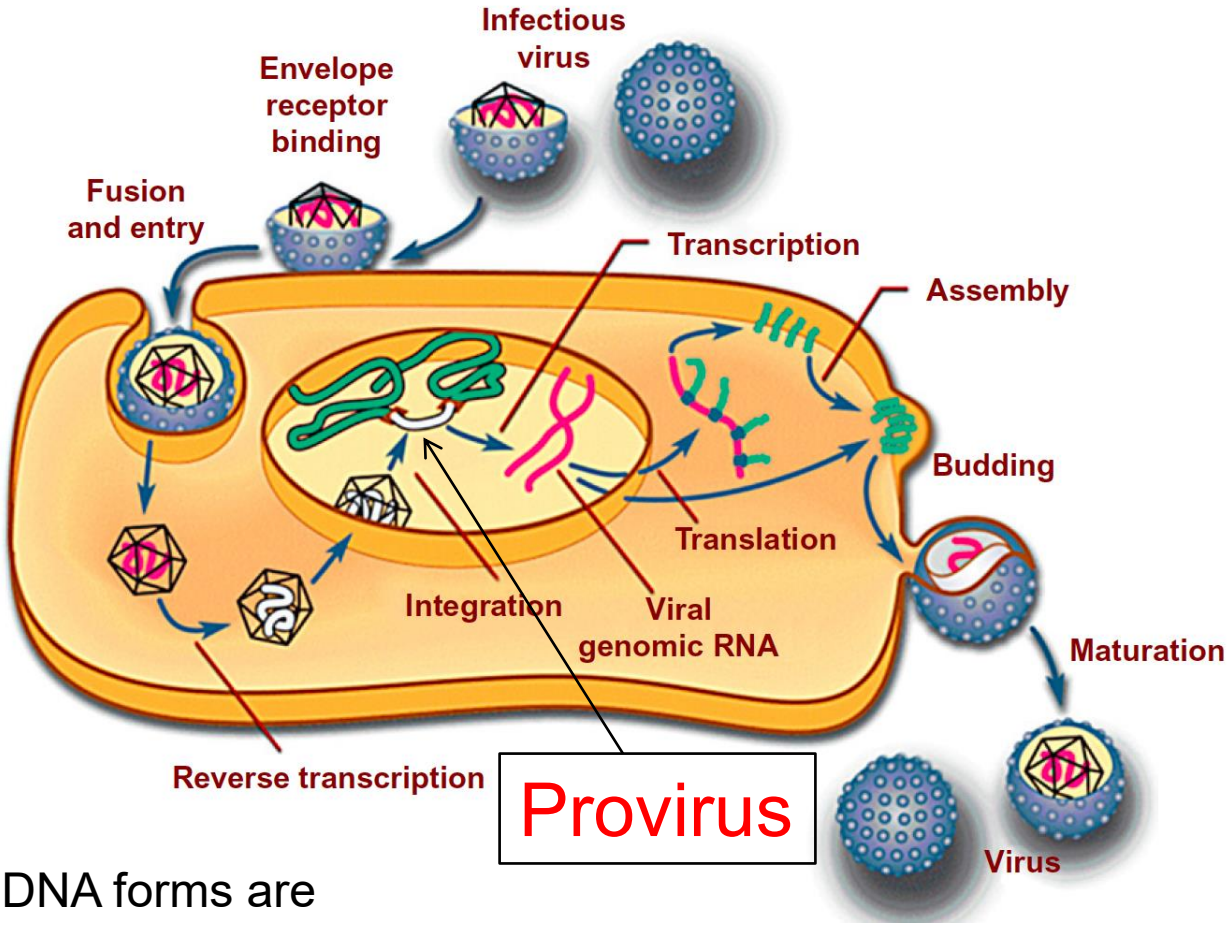


Fenner's Vet. Virol. (5th Ed.)

Viral proteins

- *gag* (group-specific antigen; gs antigen)
  - p27 (27 kDa)，為capsid antigen (CA)。
- *pol* (polymerase)
  - Reverse transcriptase (RT)
  - Integrase (IN)，為病毒DNA嵌入宿主DNA所需。
- *env* (envelope)
  - gp85 (SU)，receptor binding glycoprotein，為棘的頂部的節狀構造，為決定亞群的蛋白質
  - gp37 (TM) 為封套蛋白穿過外膜的區段。

Virus infection and replication



Once these DNA forms are integrated into the host cell chromosome, the retroviral genomic copies are designated as “proviruses”

Incubation Period

- virus strain, route, age at exposure, genetic constitution of the host
- virus strain lacks or possesses a viral oncogene
  - slowly transforming
    - viruses that do not carry a viral oncogene
    - activate cellular proto-oncogene: LTR promoter/enhancer 功能，可驅動宿主細胞進行不正常轉錄，使細胞癌化
    - non-defective
    - It is very seldom that LL cases occur in chickens under 14 weeks
  - acutely transforming
    - Viruses that have a viral oncogene
    - Insert oncogene into target cell genome
    - defective

Virus class according to envelope subgroup

subgroup	Occurrence		Oncogenicity
A	common	exogenous	+
B	less	exogenous	+
C	rare	exogenous	+
D	rare	exogenous	+
E	common	endogenous	-
J	common	exogenous	+

ALV-K??

China

2012: JS11C1, JS11C2, and JS11C3

2014: JS13-DX5, JS13-LH1, GD14LZ, GDFX0601 GDFX0602, GDFX0603

2015: GD150509

2016: GD160403, GD160607

2019: HB2015032



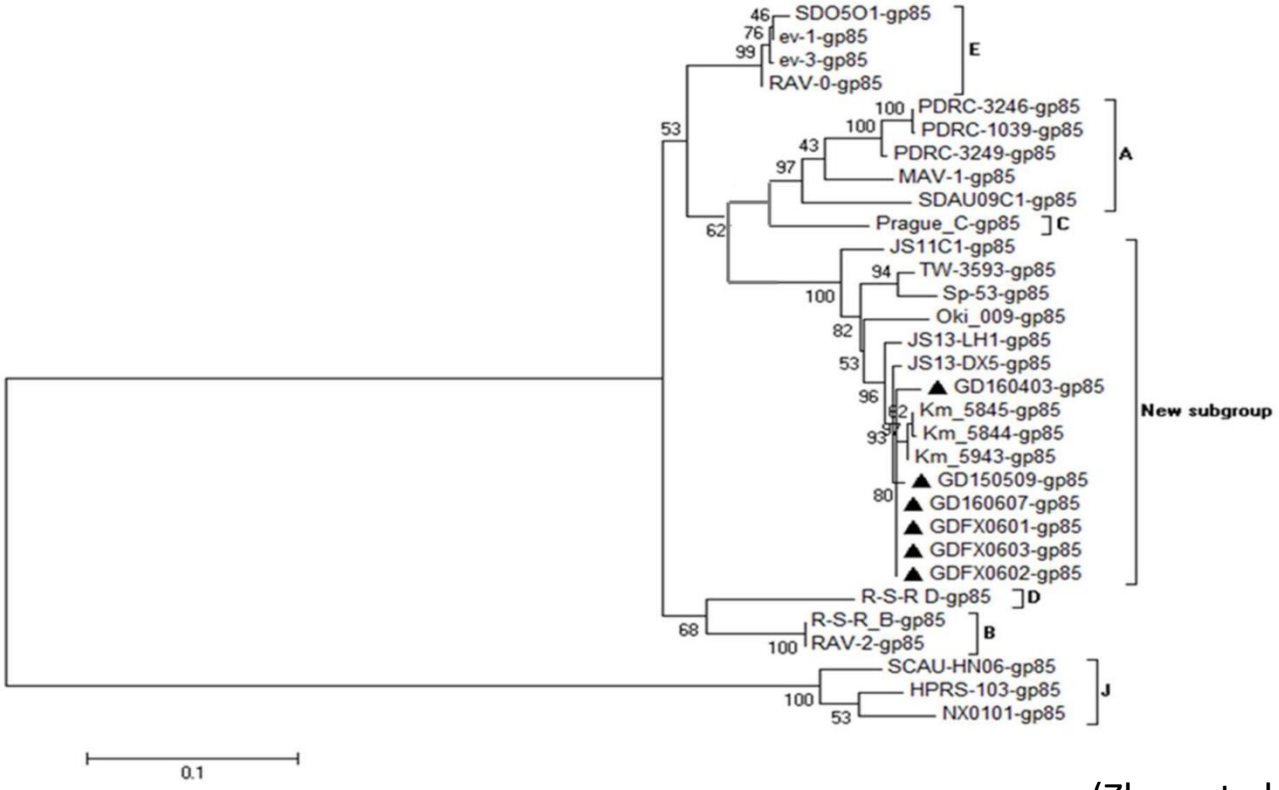
Japan, 2012

Sp-53, Oki_009, Km_5844, Km_5943, Km_5845

Taiwan, 2009

TW-3593

ALV-K??



(Zhao et al., 2018)

- Strong interference between subgroup A and K

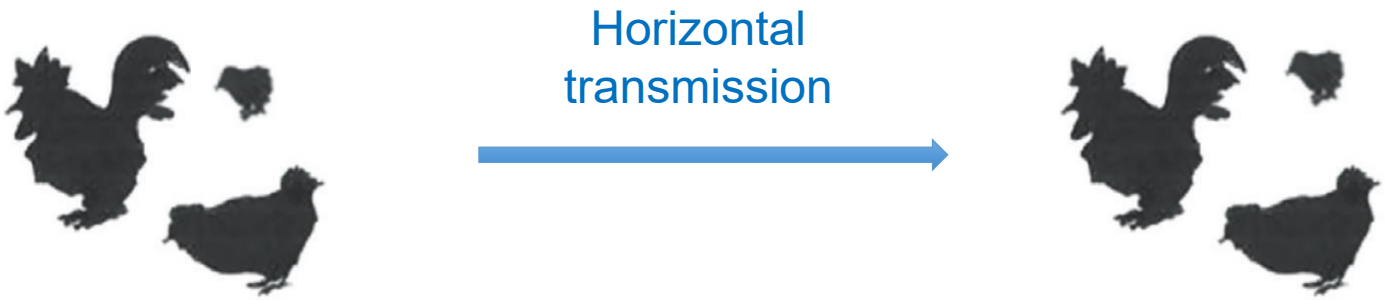
(Přikryl et al.,2019)

Host

- Subgroup A, B, C, D, E, J：Chickens
- Subgroup F, G, H, I：內源性；雉雞，松雞，鵪鶉
- 人工感染在許多禽類可以成立，如 Rous Sarcoma Virus 可在雞、雉雞、珠雞、鴨、鴿子、鵪鶉、火雞等形成腫瘤。

Transmission

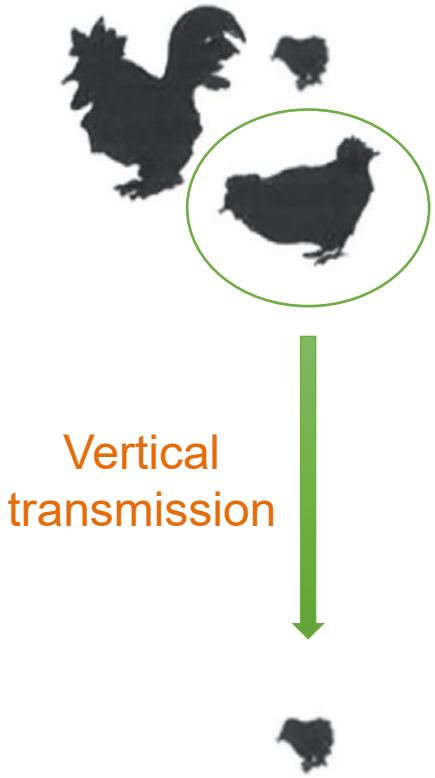
Exogenous: Horizontal transmission



- direct (bird to bird)
  - ALV-J > ALV-A
- indirect contact (environment) : rare
- oculonasal, oral, respiratory

Transmission

Exogenous: Vertical transmission



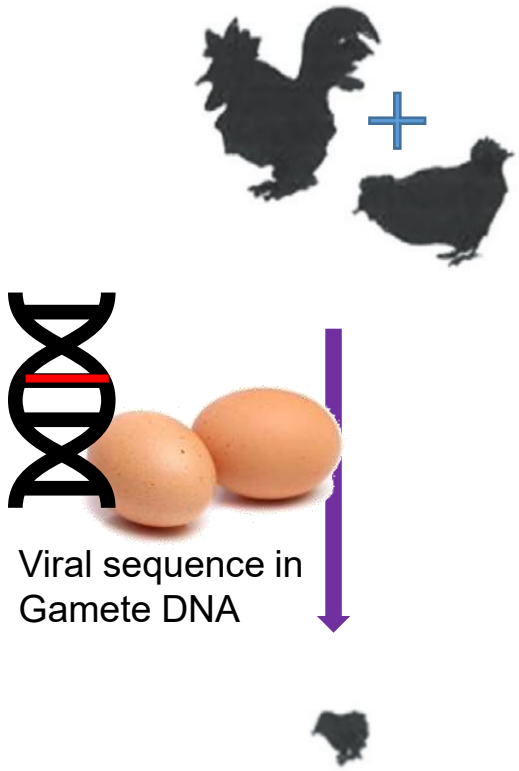
- congenital transmission 先天傳染
- female
  - from oviduct to egg
- immunologic tolerance 免疫耐性
  - persistent viremia
  - no obvious antibody response
- keep shedding large amounts of virus

免疫耐受性 (Immune tolerance)

- 垂直感染造成
- 不會產生抗體
- 終生帶毒及排毒
- 可形成腫瘤

Transmission

Endogenous: Genetic transmission



- genetic transmission
- female and male
- generally no disease (latent)
- may influence the response of the bird to infection by exogenous ALVs by inducing immunological tolerance or immunity
- EAV-HP: source of the envelope gene of ALV-J

ALV-A, B transmission



Horizontal

Rooster ♂ / Hens ♀



Vertical (Hens ♀)

V- A- : resistance
V+ A+ : rare
V- A+ : >80%
V+ A- : <10%, LL

Vertical shedder



Horizontal



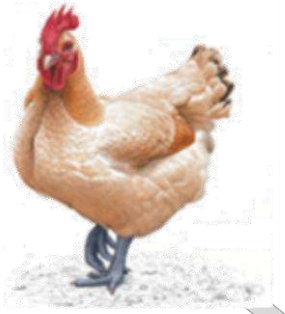
V+ A- **S+**
Lymphoid Leukosis
Immune tolerance

V- **A+** S-

Viremia
Antibody
Shedding

ALV-J transmission

V- A+ **S+**
V+ A- **S+**
Hens ♀



Viremia
Antibody
Shedding

Vertical



V+ A- **S+**



Early
Horizontal

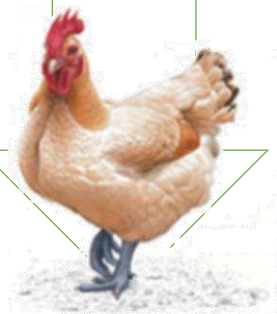


V+ A- **S+**
V- A+ **S+**

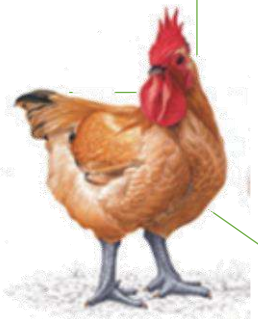


V- A+ **S-**

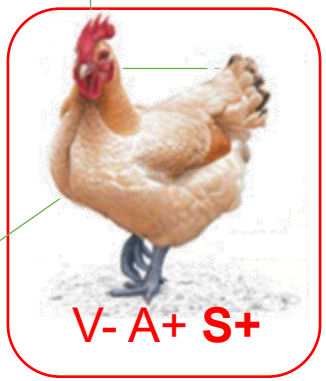
Late
Horizontal



V+ A- **S+**
Myeloid Leukosis



V+ A- **S+**
Myeloid Leukosis



V- A+ **S+**

Virus transmission

subgroup	Route	Age	Pattern	Oncotype
A/B	Vertical	Egg	V+ A- S+	Tolerance LL
	Horizontal	Chick	V- A+ S-	-
	Horizontal	**Mature**	**V+ A- S+**	**<10%, LL**
	Horizontal	Mature	V- A+ S-	>80%
E	**Genetic Vertical**	**Gamete**	Latent	
J	Vertical	Egg	V+ A- S+	ML
	Horizontal	**Chick**	**V+ A- S+**	ML
	Horizontal	**Chick**	**V- A+ S+**	-
	Horizontal	Mature	V- A+ S-	-

Clinical Signs

- non-specific
  - Inappetence (食慾不振), weakness, diarrhea, dehydration, emaciation(消瘦), pale wattles(雞冠蒼白)
- abdominal enlargement

Neoplastic diseases

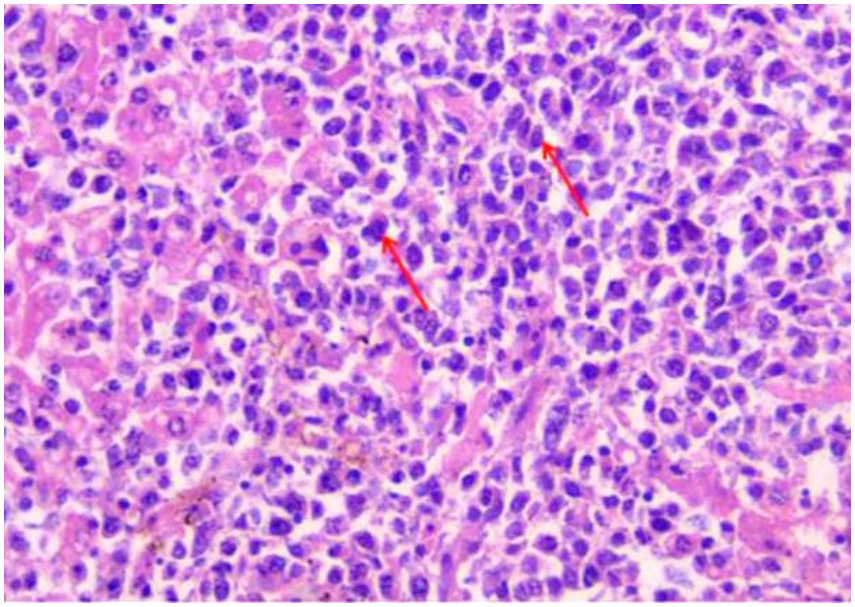
- **Lymphoid leukosis (LL)**
- Myelocytomatosis/ **Myeloid leucosis (ML)**
- Erythroblastosis
- Myeloblastosis
- Sarcomas

病變

- 一般至17週就可發生腫瘤，大多為骨髓細胞瘤或骨髓母細胞瘤。肉雞最早於9週齡首見腫瘤，出現腫瘤的中位數為20週齡。
- 感染急性轉型毒株，最早於感染後5週齡出現腫瘤，中位數為9週齡，因其病毒可能含致癌基因。每月死亡率可達6%以上，死亡雞隻有60-70%可見不同腫瘤。
- 最常見的腫瘤為淋巴球性白血病，為ALV-A所引起的，除了內臟腫瘤以外，有時可以見到華氏囊結節性腫大。
- ALV-J引起多發骨髓球性白血病 (myeloid leukosis)。

Lymphoid Leukosis (LL)

- ALV-A
- older than **4** months
- Abdominal distention



Lymphoid leukosis



Pathological findings-LL

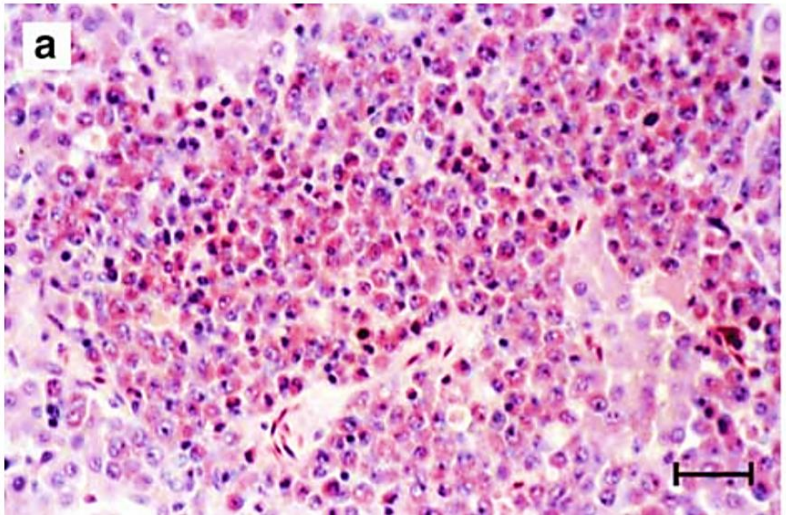
- liver, spleen, kidneys, heart, ovary, bursa
- diffuse or nodular

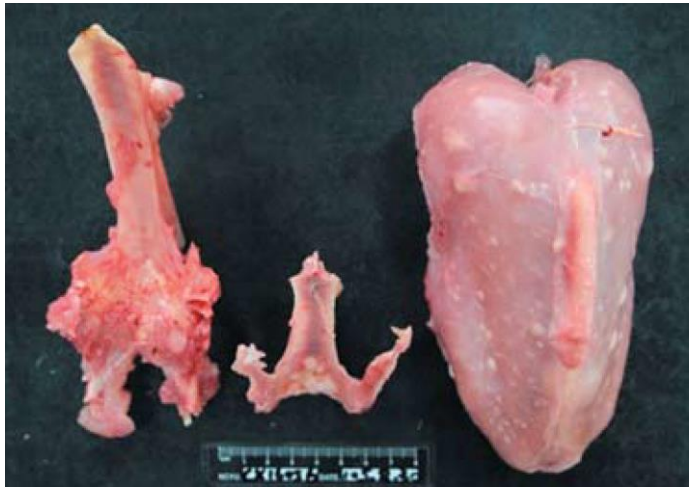


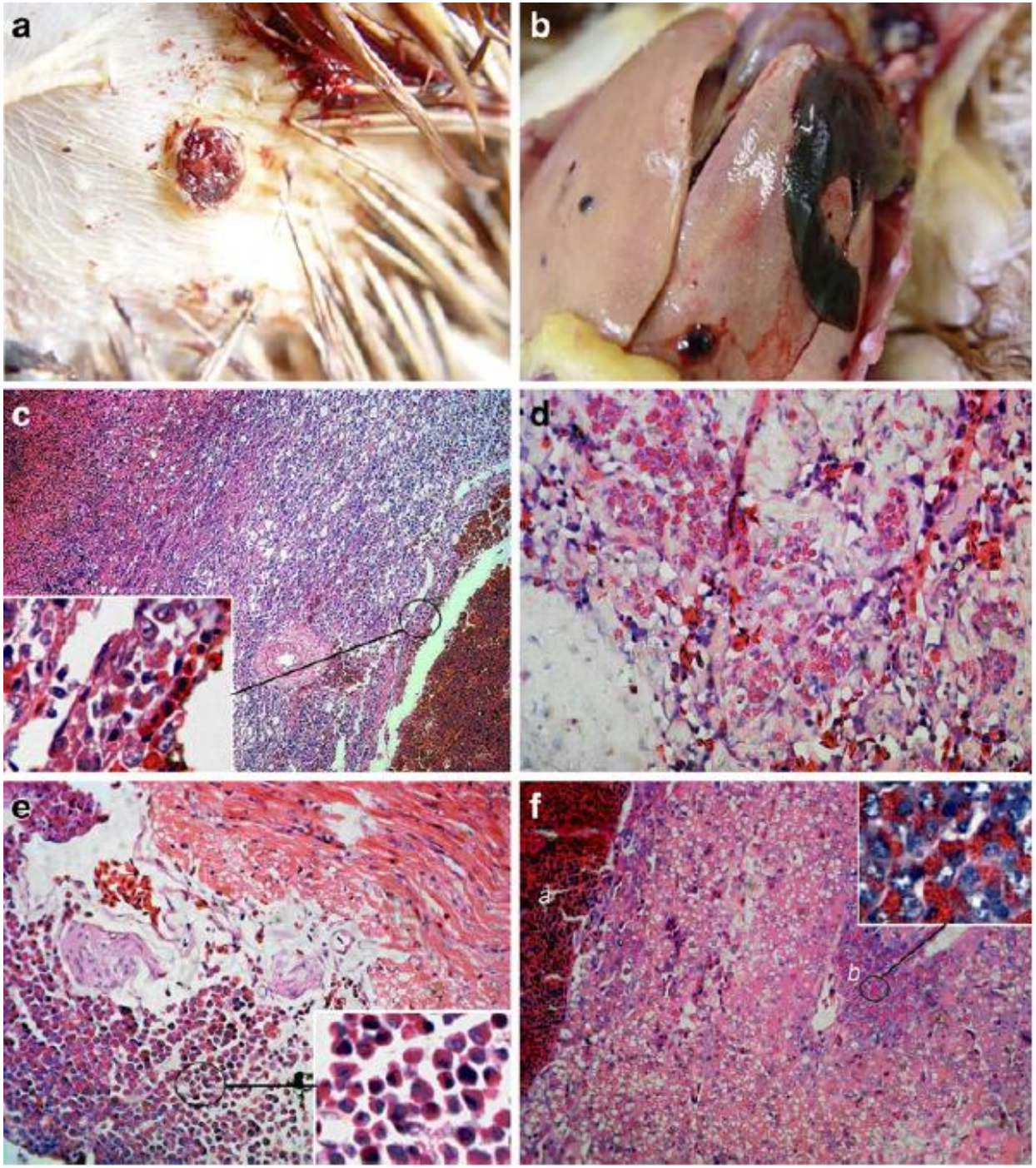
Myeloid Leukosis (ML)

- ALV-J
  - Acute: 4w
  - Chronic: 20w
- liver and spleen enlarged








a
b
c
d
e
f
a
b







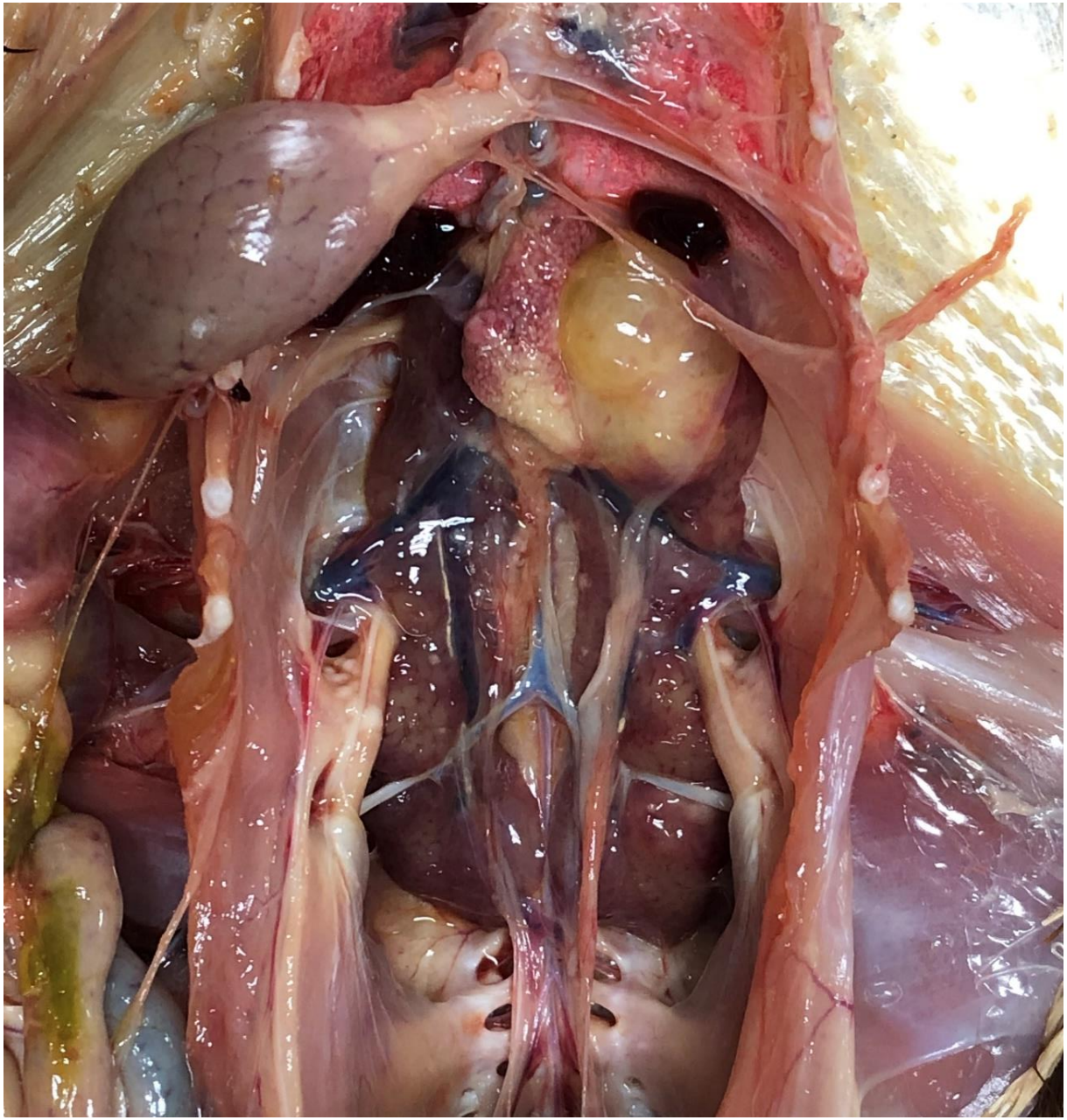


Li et al., 2021

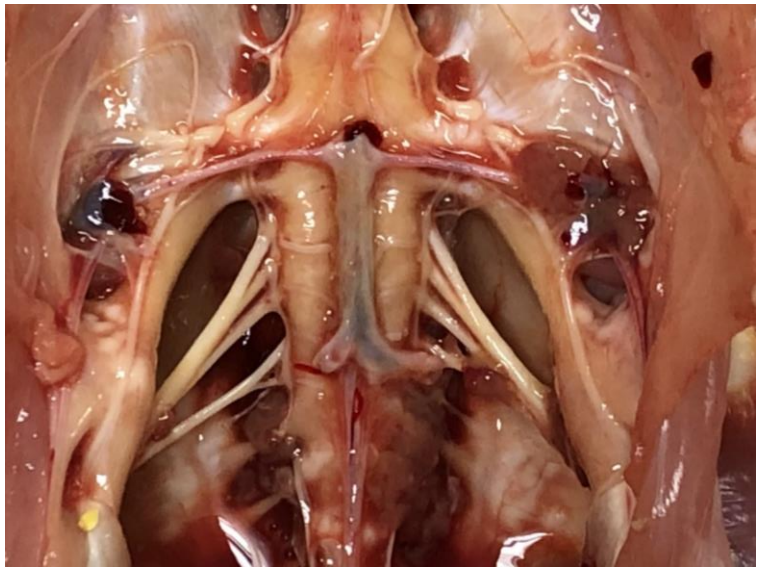

NTUVP
臺大獸醫病理



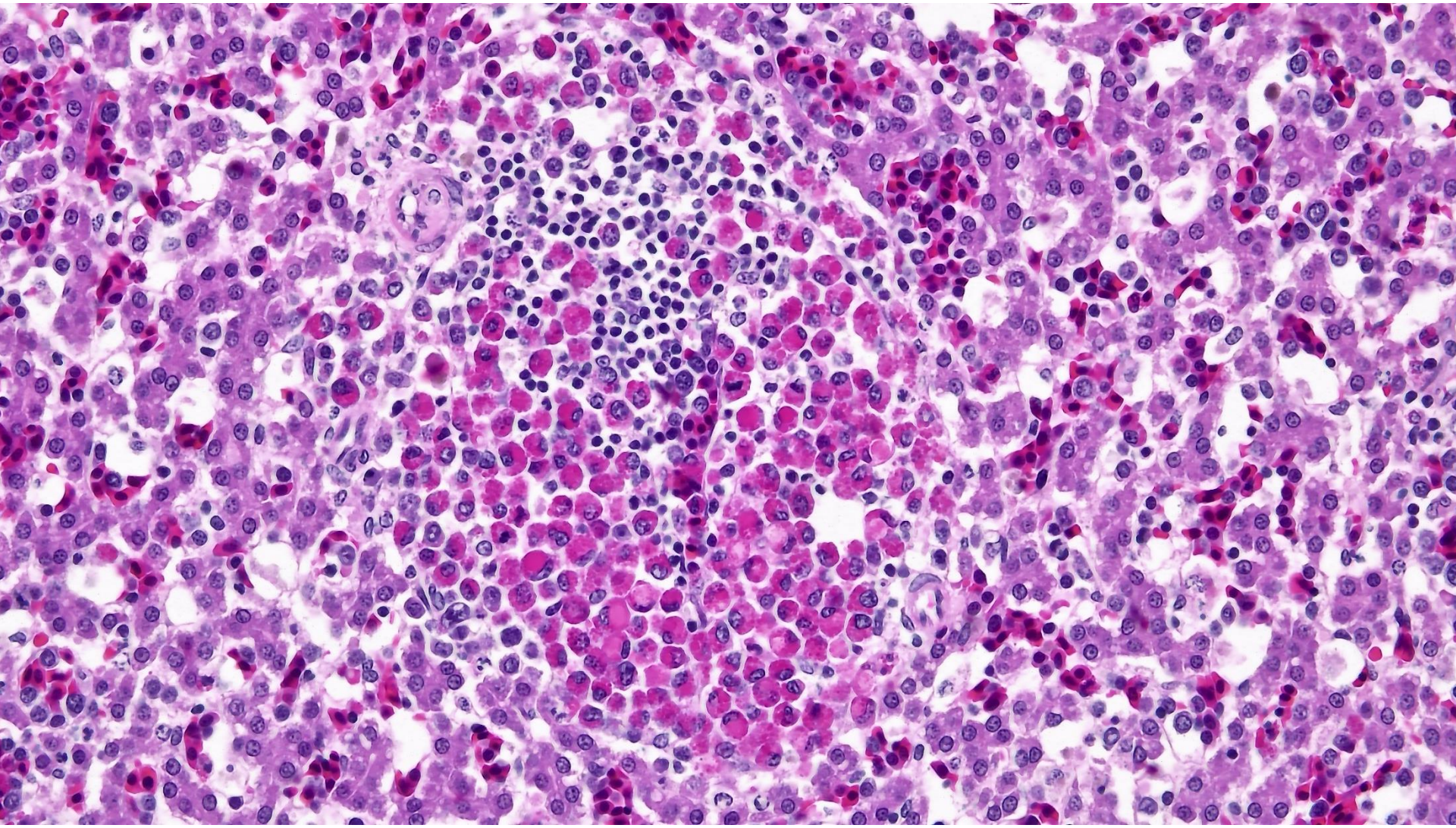


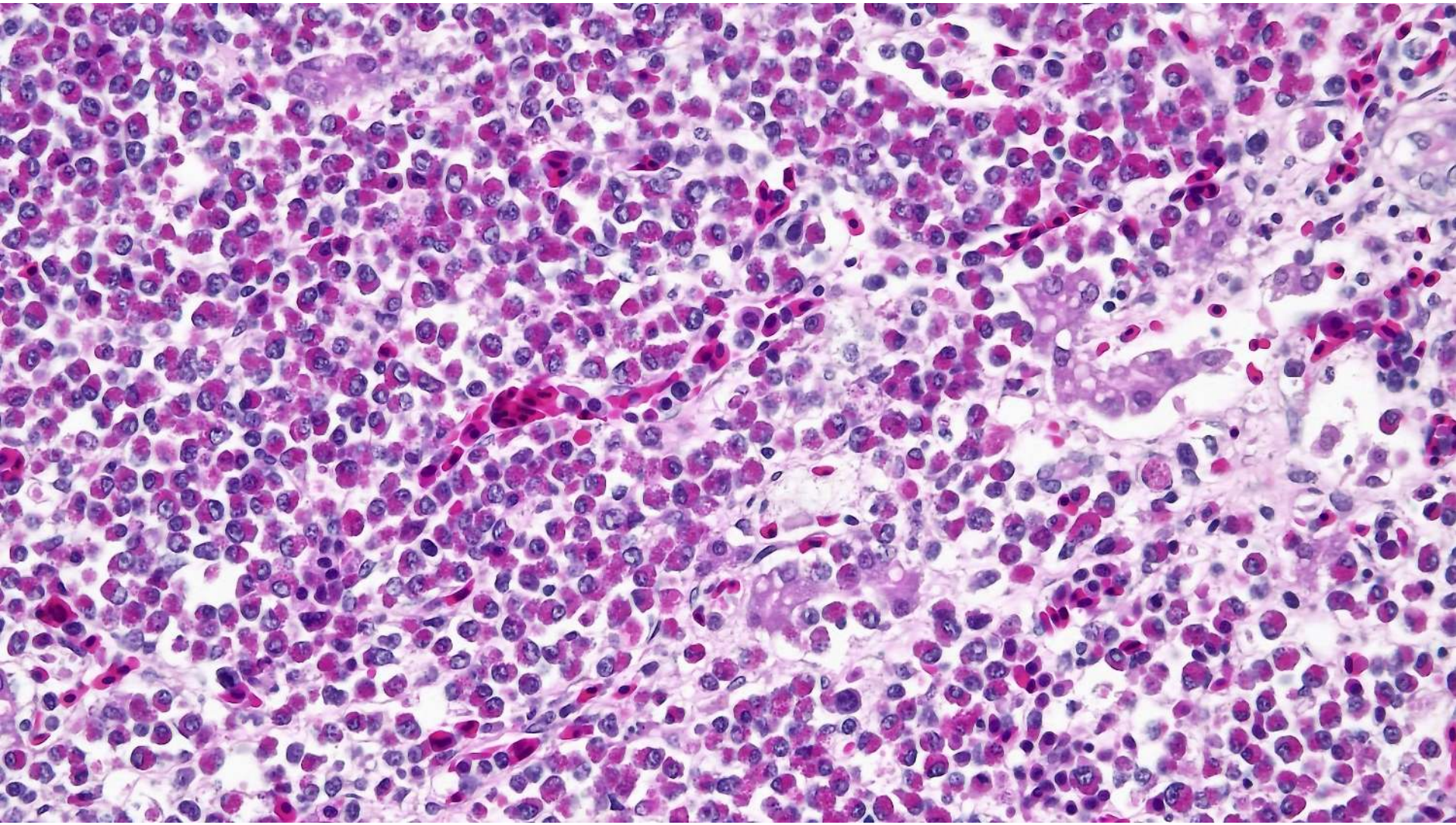


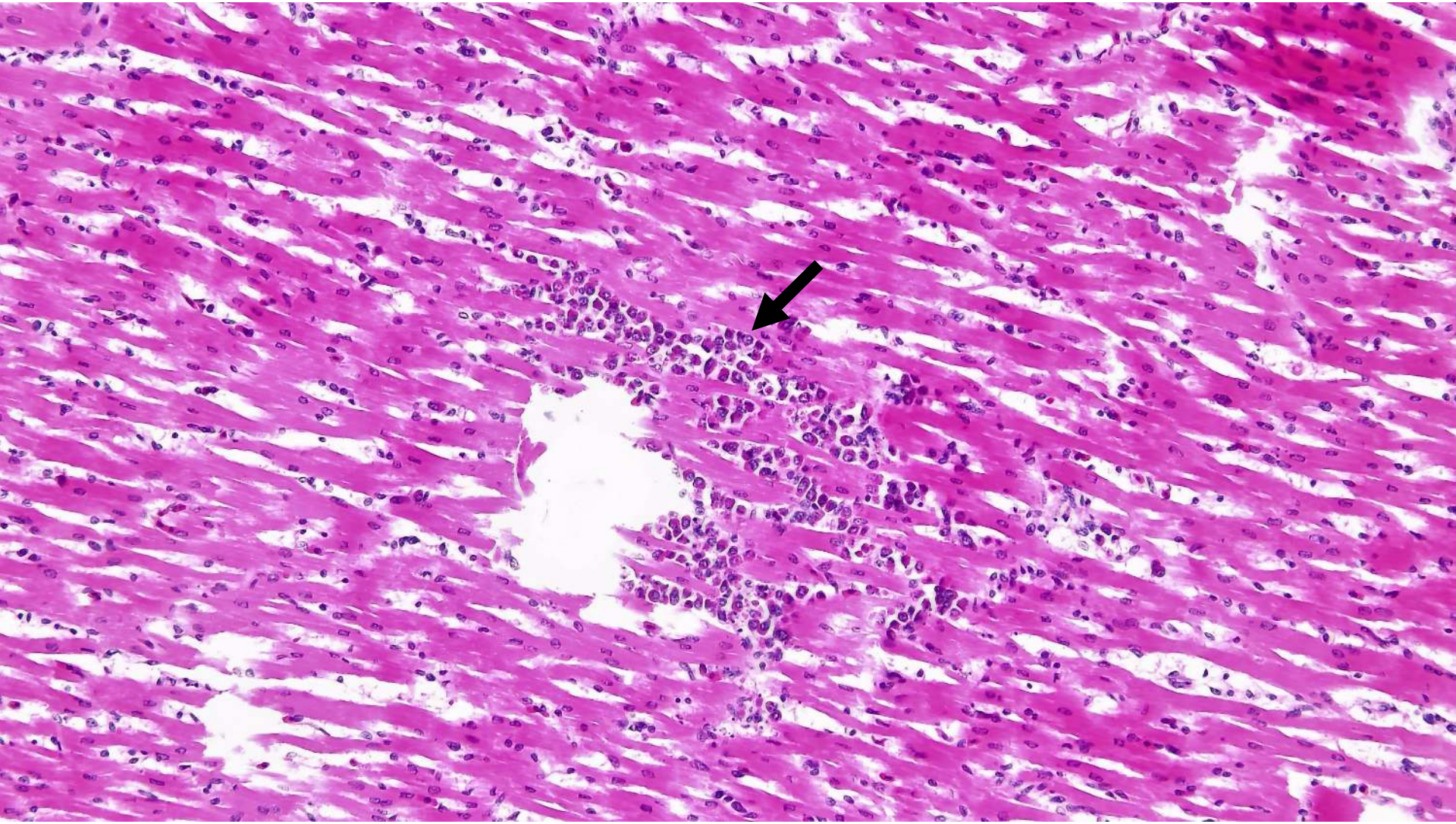

NTUVP
臺大獸醫病理
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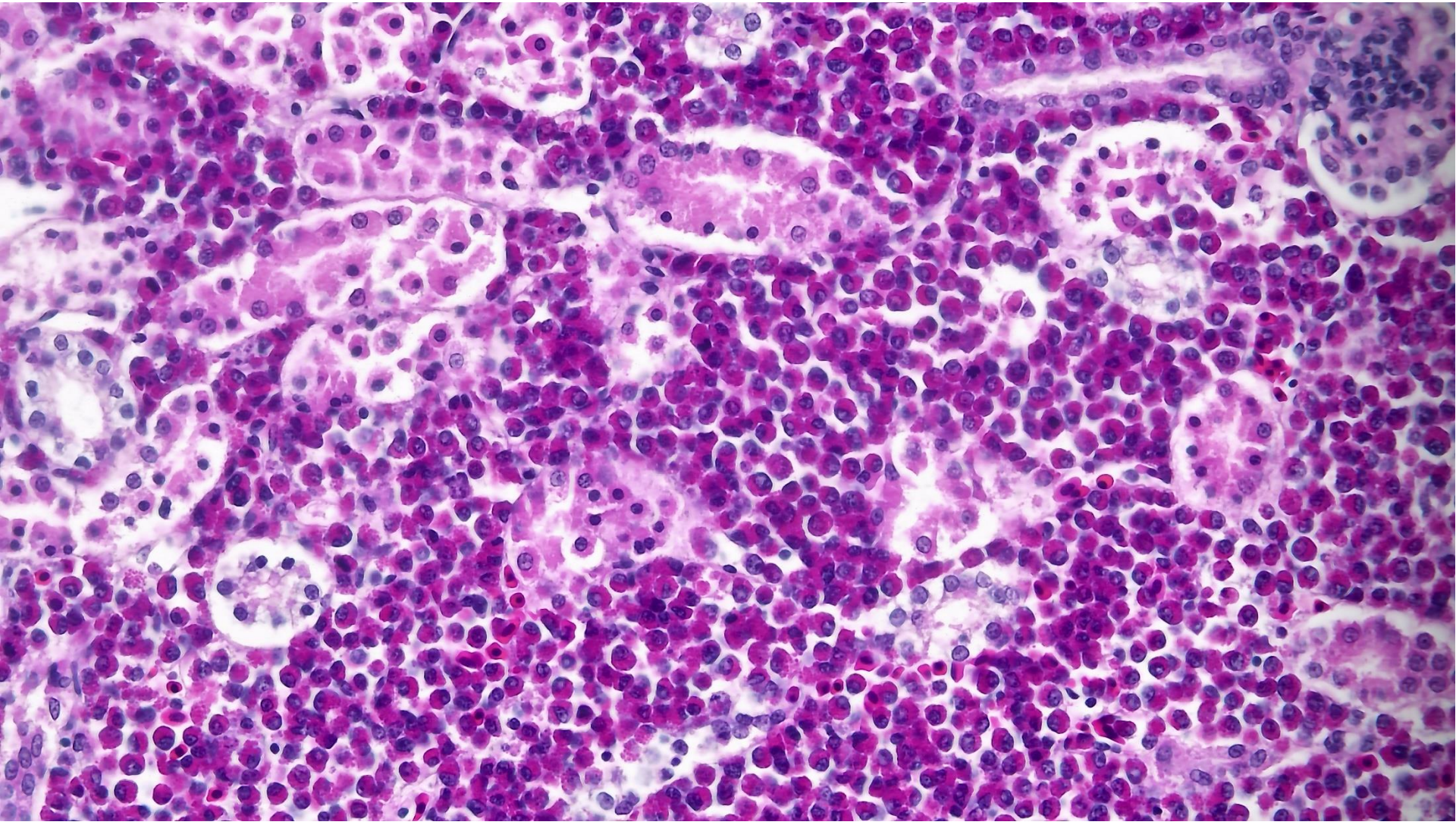






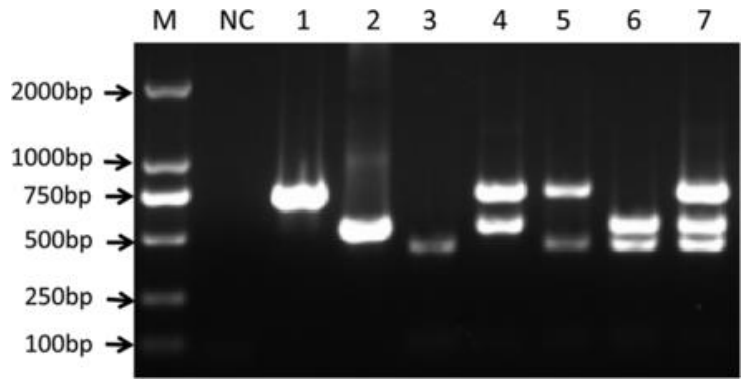






Laboratory diagnosis

- Virus isolation
  - gold standard
- PCR (provirus)/RT-PCR (virus)
  - multiplex PCR: MDV, ALV, REV
  - ALV subgroup
- ELISA (Ag)
  - commercial kit: detect p27
  - detect both exogenous and endogenous viruses
  - can’t discriminate between different ALV subgroup



(Gao *et al.*, 2013)

其他診斷的困難

- 可以進行檢測的實驗室不多
- 病毒分離不易且耗時
- 無所不在的內源病毒干擾
- 無細胞病變 (cytopathic effect)
- 需使用對ALV-E有抵抗力的細胞株，如DF-1

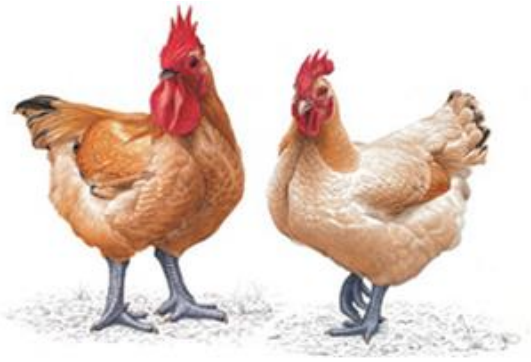
Treatment, Prevention and Control

- no specific treatments
- no vaccine
- eradication
- selection for genetic resistance

Eradication of exogenous ALV

- 1977, 商品化雞場已清除LL
- 1991, 英國國家獸醫研究院分離出一種新型的家禽白血病病毒 (HPRS-103株)，此病毒與之前亞型皆不相同，而被歸為J亞群，稱ALV-J
- 1996-2002, 全球發生ALV-J，許多種原被汙染
- 2004, 歐美種禽公司宣稱已清除白肉雞ALV-J
- now
  - in Europe and America:
    all GP and PS are free from ALV-A, B & J
  - in Taiwan:

	ALV-A, B &J
Broiler breeder	Free
Layers breeder	Free
Colored chicken breeder	?



種雞群ALV-J清除計畫

Age	samples	methods
1 day	meconium (胎糞)	P27 ELISA, RT-PCR, virus isolation
6 wks	plasma, cloacal swab	RT-PCR, PCR, virus isolation
18 wks	plasma, cloacal swab	RT-PCR, PCR, virus isolation
22 wks	plasma, albumin, cloacal or vaginal swab	RT-PCR, PCR, virus isolation
25-30 wks	plasma, cloacal or vaginal swab	RT-PCR, PCR, virus isolation
45 wks	plasma, cloacal or vaginal swab	RT-PCR, PCR, virus isolation

建議處置

- 治療及控制:
  - 家禽白血病無有效的治療方式
  - 家禽白血病無疫苗
  - 建議該場淘汰發病及疑似發病雞隻
  - 適當補充維生素、電解質以降低緊迫之影響

建議處置

- 消毒:
  - 勤加清掃禽舍減少灰塵與毛屑
  - 雞舍空欄後，全面以高壓水柱清洗再以過氧化物類或鹵素類消毒劑消毒
  - 消毒後空欄至少四週以上再引進雞隻

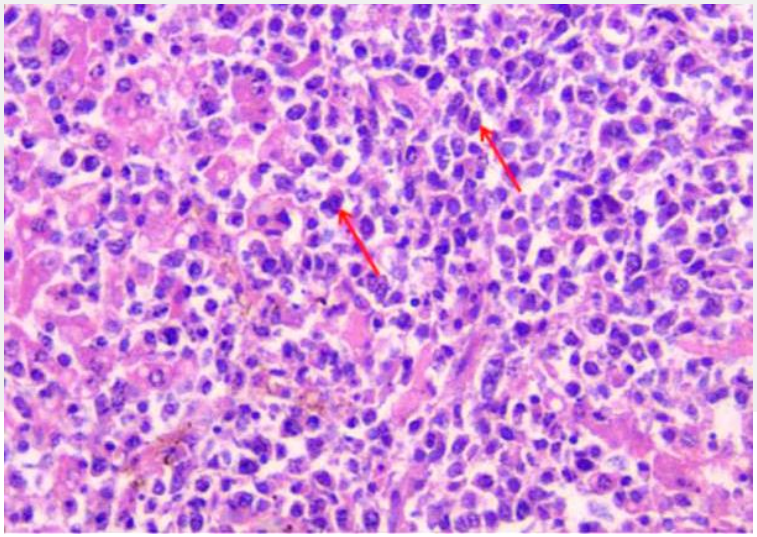
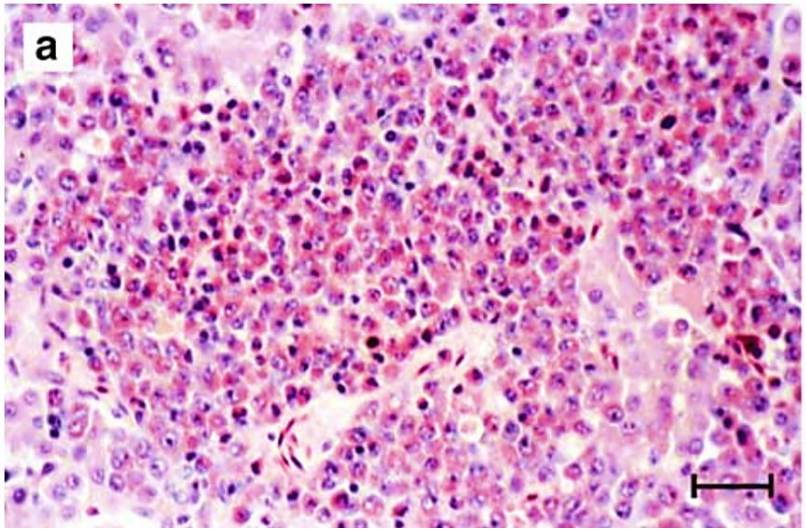
建議處置

- 飼養管理:
  - 加強生物安全措施，加強人車出入口之消毒及管理
  - 考慮經檢測選購家禽白血病陰性之種雞
  - 確實做好馬立克病之免疫注射

Summary

subgroup		Virus transmission	Oncogenicity
A	exogenous	• Vertical (by female) • Horizontal	+ (LL)
E	endogenous	• Genetic	-
J	exogenous	• Vertical (by female) • Horizontal	+ (ML)

Take home message

家禽白血病A亞群 (ALV-A)	家禽白血病J亞群 (ALV-J)
淋巴球性白血病 Lymphoid Leukosis (LL)	骨髓細胞白血病 Myeloid Leukosis (ML)
	

Reticuloendotheliosis (RE)

History

- 最早出現的家禽網狀內皮增生病毒（Avian Reticuloendotheliosis Virus, REV）於火雞體內之淋巴瘤臟器分離得，在經過火雞及雞隻體內連續繼代 300 代後所得之分離株為 T strain (Robinson and Twiehaus, 1974)
- 台灣於1987年由陳和謝首次發表白羅曼鵝病例

REV in Taiwan

	Species	Results of Ab detection	Results of PCR detection	
1987	chickens	17.5%	—	Chen and Shieh,1987
	geese	37.97%	—	
	ducks	34.72%	—	
2003-2005	chickens (breeder)	92.8%	—	Cheng *et al*., 2006
	chickens (colored breeder)	8%	29%	
2008	geese	60.3%	28.8%	Lin, 2008

Avian reticuloendotheliosis virus (REV)

- Classification:

  Retroviridae

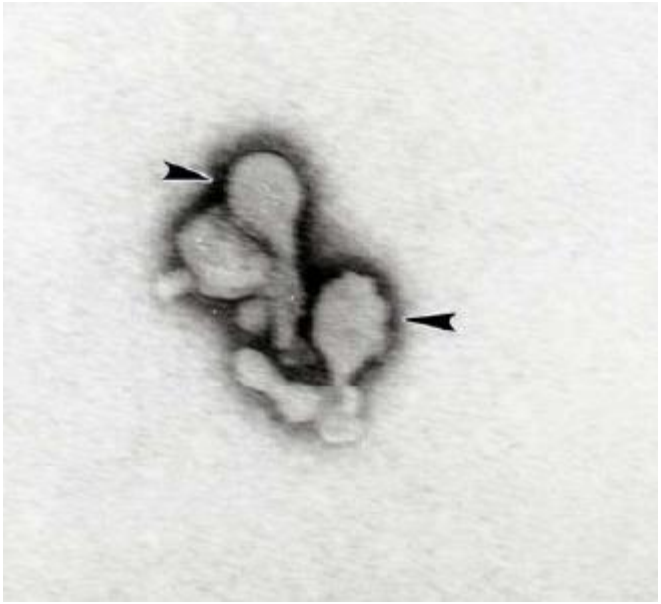
  Gammaretrovirus

  Morphology:

  enveloped

  diameter: 100 nm

  ssRNA(+)



(Cheng, 2005)

REV groups:

- replication defective virus strain T
- nondefective strain A
- chick syncytial virus (CSV)
- spleen necrosis virus (SNV)
- duck infectious anemia virus (DIA)

雖然各不同毒株抗原性有差異，但都為同一群，無不同亞群存在

Host

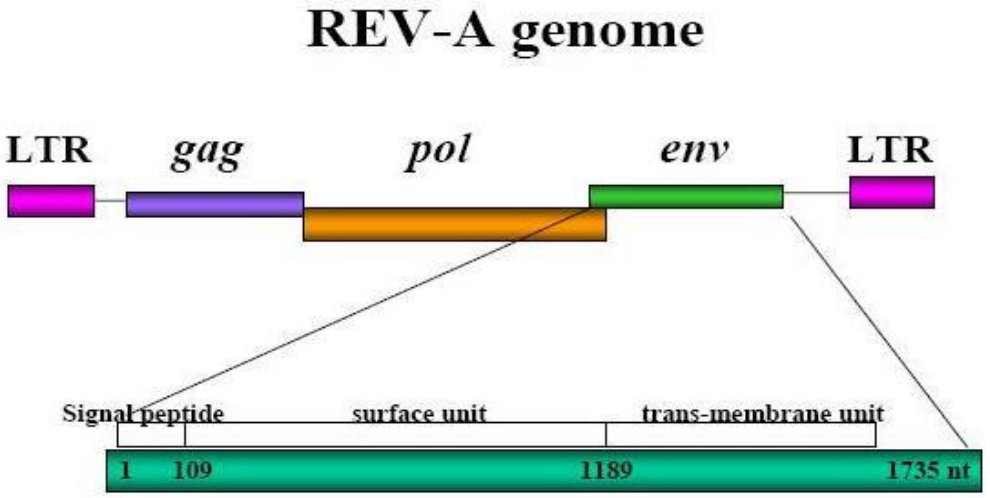
- 雞
- 火雞
- 鴨
- 孔雀
- 鵝
- 雉雞
- 珠雞
- 鵪鶉
- 松雞





REV A strains

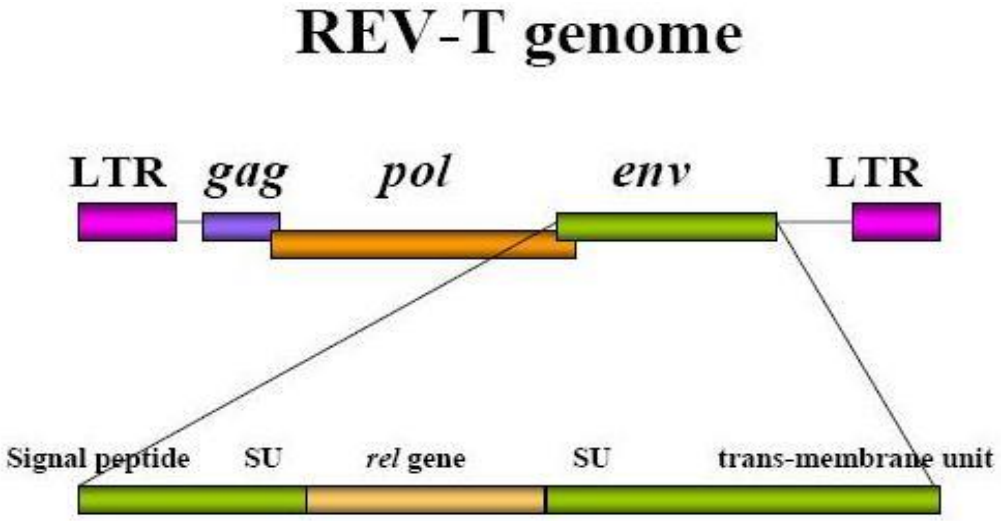
- Non-defective strain
- helper virus for strain T
- 複製完整型，不具急性致癌能力
- 慢性腫瘤
- runting disease (矮小症候群)



9 kb

REV T strains

- replication defective
- 具急性致癌基因*v-rel*
- 需要helper virus (REV A strain)幫忙複製
- 急性網狀細胞腫瘤
- 在CEF或DF-1培養
  會喪失癌化能力
  →繼代三代後:
  REV-T 消失
  REV-A 存在

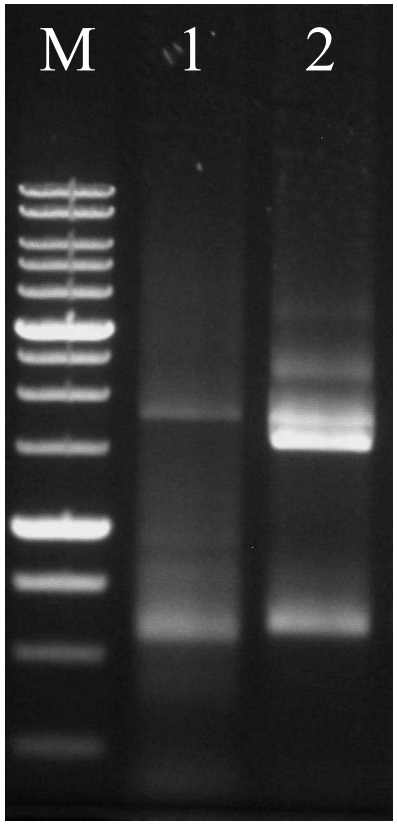


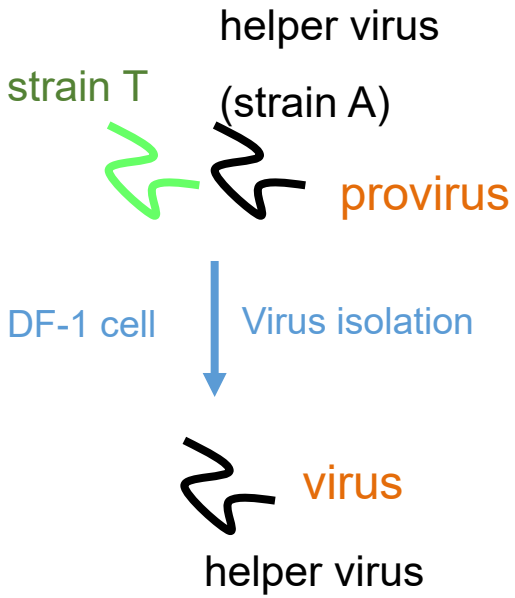
5.7 kb

3472 PCR/RT-PCR product

Lane M: 1 kb marker
Lane 1: RT-PCR product (virus)
Lane 2: PCR product (provirus)

Primer: env5/mRenR





傳播

- 直接接觸傳染。
- 急性感染期由糞便排毒、污染墊料。
- 昆蟲、蚊子機械性傳播。
- 介蛋傳染、但機率不高。
- 疫苗污染：禽痘疫苗、馬立克病疫苗。

REV污染疫苗

- 禽痘疫苗及馬立克病疫苗時常發現有REV基因的插入。
- REV基因插入禽痘疫苗可分為3種：全長基因、長LTR及短LTR。僅插入LTR片段的疫苗不被認為污染，因其不會造成臨床發病，其接種的雞隻也不會產生對抗REV的抗體。
- (在美國) 大多數的禽痘野外毒含有全段的REV基因，其病原性較高，而疫苗大都含有不同長度的REV基因片段。

Incubation Period

- virus strain
- Acute: 3 days
- 慢性腫瘤:
  - 雞 17-43 wks
  - 火雞 8-12 wks
  - 鴨 4-24 wks
  - 鵝 20-30 wks

Clinical Signs

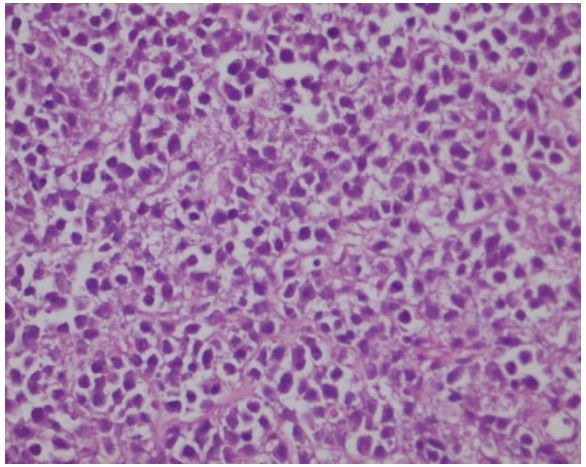
- 常無明顯症狀
- runting disease syndrome
- 羽毛生長不良
- 免疫抑制

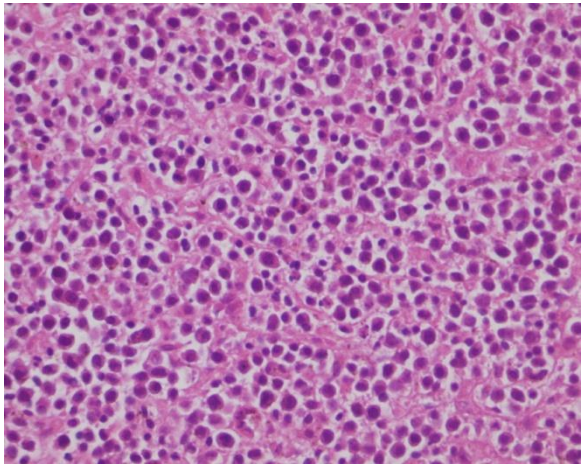


Case 3461

- mule duck
- Taichung, Taiwan, 2007
- 6 / 4-5 month-old
- showing no obvious clinical signs









Case 3471

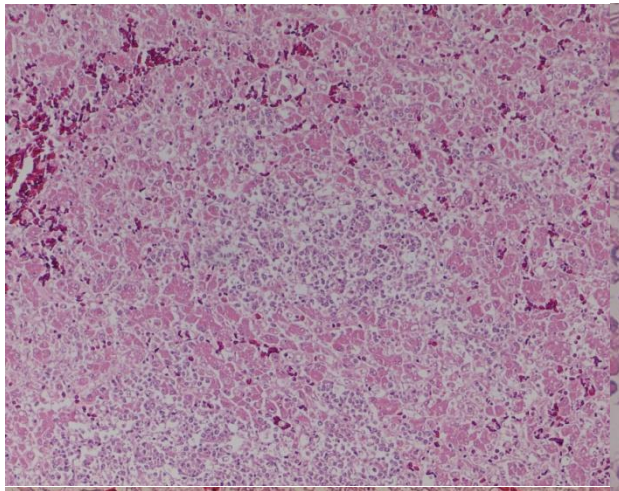
- white Roman goose
- Yunlin, Taiwan, 2008
- 70 day-old
- listed below weight



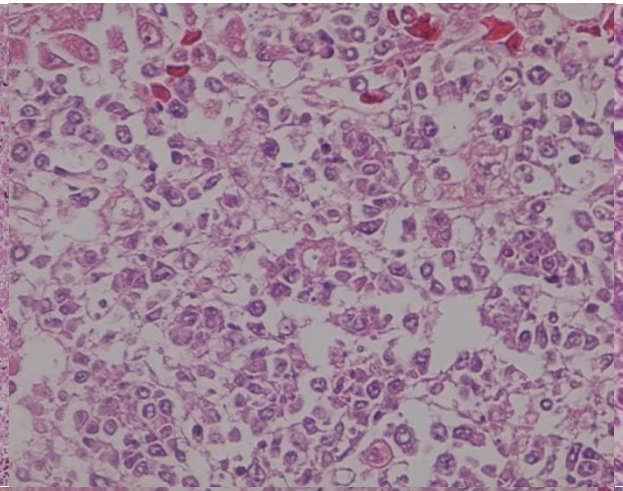


(Photographs were provided by Dr. Jui-Hung Shien)

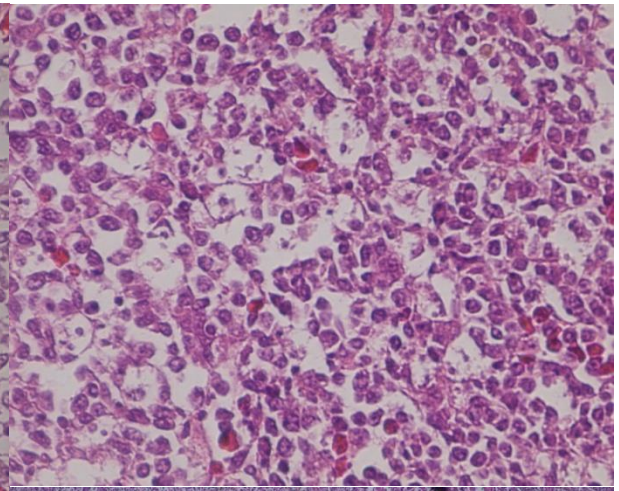
(Liver, H&E, 100X)



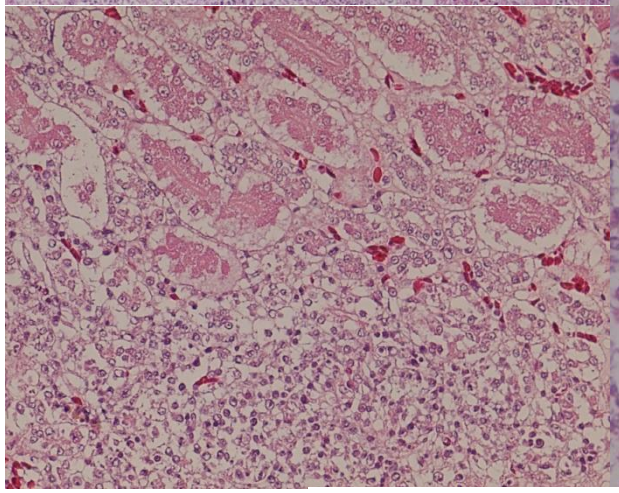
(Liver, H&E, 400X)



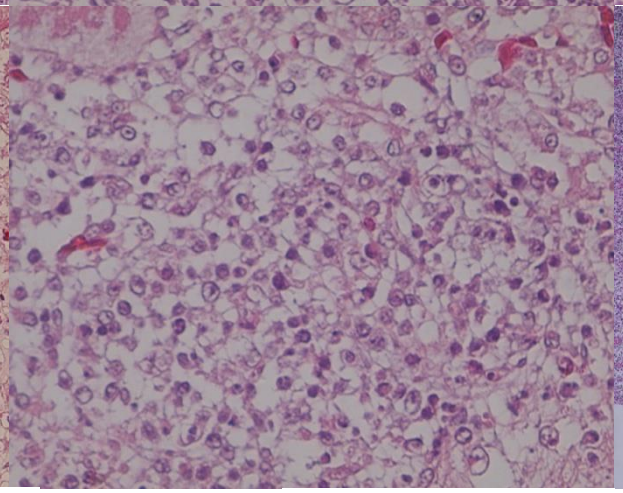
(Spleen, H&E, 400X)



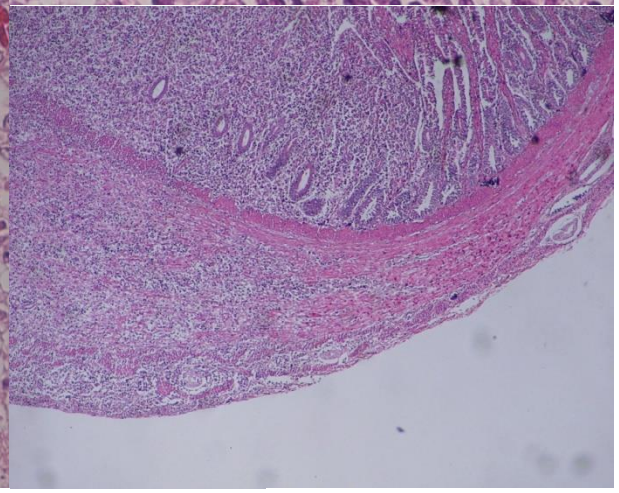
(Kidney, H&E, 100X)



(Kidney, H&E, 400X)

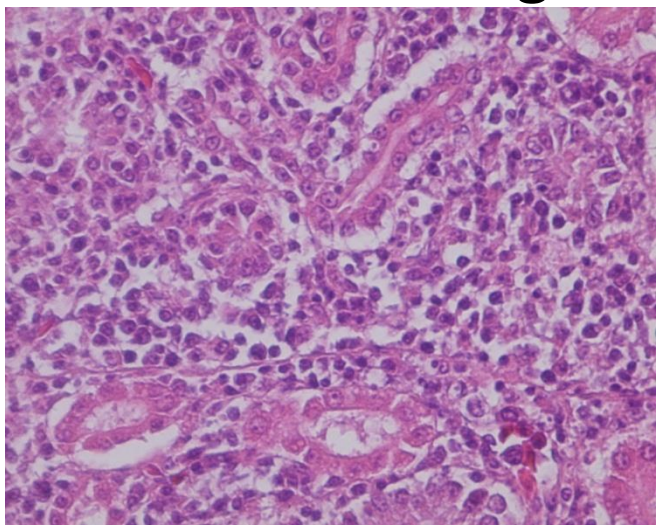


(Intestine, H&E, 40X)



Case 3472

- white Roman goose
- Yunlin, Taiwan, 2008
- 70 day-old
- listed below weight



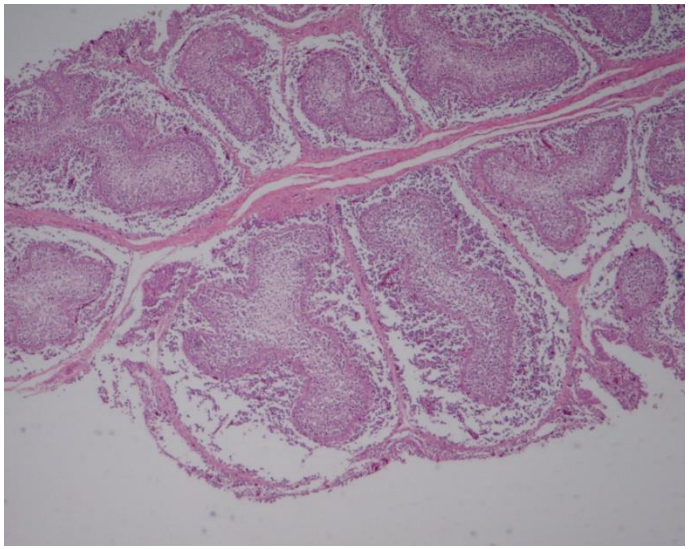


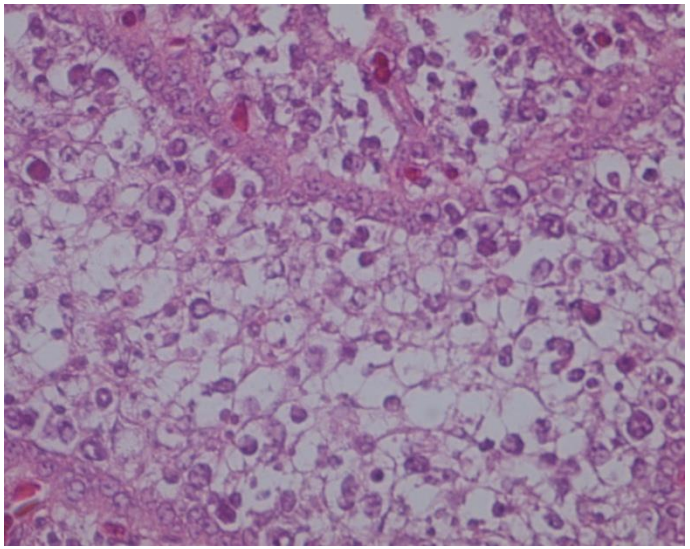
Case 3473

- white Roman goose
- Yunlin, Taiwan, 2008
- 70 day-old
- listed below weight



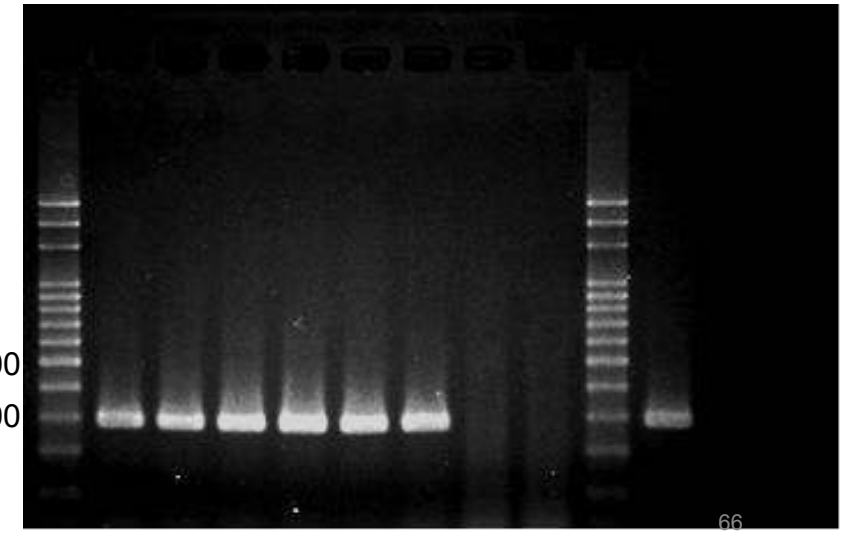






Laboratory diagnosis

- Virus isolation
  - buffy coat, tumor tissue
    - → DF-1 / CEF (REV-T strain會消失)
- PCR (provirus)/RT-PCR (virus)
  - multiplex PCR: MDV, ALV, REV



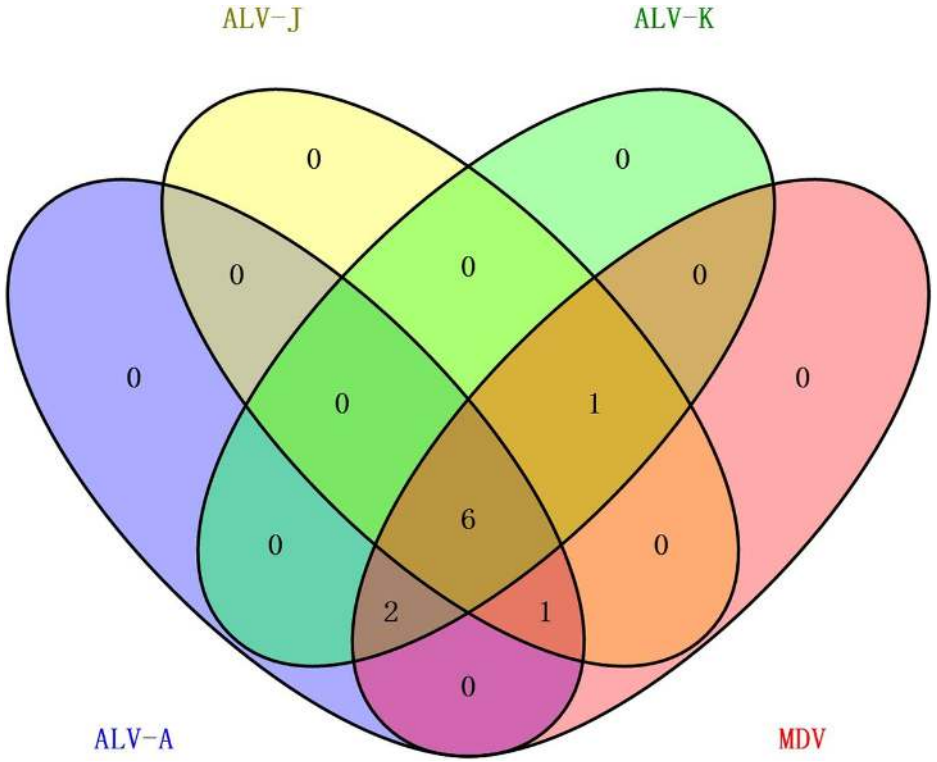
Laboratory diagnosis

- IFA
- IHC
- ELISA
  - Antigen
  - Antibody

Treatment, Prevention and Control

- no specific treatments
- no vaccine
- 生物安全
- 疫苗廠商應隨時監測雞胚胎或細胞不受REV的污染
- eradication

- Co-infection of MDV with ALV-A, J and K
  - 肝臟腫瘤
  - 組織切片:MDV、LL
  - PCR:



(Li et al., 2019)

- Co-infection of MDV and REV
  - Approximately 50% of detected tumor chickens in Chinese flocks were GaHV-2 positive
  - Co-infection of REV occurred in 13% of GaHV-2-positive samples.
  - Co-infection increased mortality and tumor formation in experimental infection
  - REV/GaHV-2 co-infection may contribute to persistent neoplastic disease in poultry

(zhang et al., 2019)

Differential Diagnosis

	MD	LL	ML	RE
Pathogen	MDV (herpesvirus)	ALV-A (retrovirus)	ALV-J (retrovirus)	REV (retrovirus)
Host species	Chickens, Turkeys, Quail	Chickens	Chickens	Chickens, Turkeys, Quail, Geese, Ducks
Age	>4w	>14w	>14w	>14w
Transmission	Horizontal	Horizontal Vertical	Horizontal Vertical	Horizontal Vertical
neurological clinical signs	O			
nerves	O			rare
eye	O			
skin/feather follicle	O		O	O

+Laboratory diagnosis

MDV	ALV	REV
免疫抑制	免疫抑制	免疫抑制
	免疫耐受	